Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $A=\left[\begin{array}{cc}\cos \theta & -\sin \theta \\ \sin \theta & \cos \theta\end{array}\right]$, then $A^T+A=I_2$, if

(A) $\theta=\mathrm{n} \pi, \mathrm{n} \in \mathrm{Z}$

(B) $\theta=(2 n+1) \frac{\pi}{2}, n \in Z$

(C) $\theta=2 \mathrm{n} \pi+\frac{\pi}{3}, \mathrm{n} \in \mathrm{Z}$

(D) None of these

Q.2

If $\mathrm{A}^{\mathrm{k}}=0($ A is nilpotent with index k), $(\mathrm{I}-\mathrm{A})^{\mathrm{p}}=\mathrm{I}+\mathrm{A}+\mathrm{A}^2+\ldots+\mathrm{A}^{\mathrm{k}-1}$, thus p is

(A) -1

(B) -2

(C) $1/2$

(D) None of these

Q.3

If A and B are two matrices such that $AB=A$ and $BA=B$, then which one of the following is correct?

(A) $\left(A^T\right)^2=A^T$

(B) $\left(A^T\right)^2=B^T$

(C) $\left(A^T\right)^2=\left(A^{-1}\right)^{-1}$

(D) None of the above

Q.4

If $\mathrm{A}=\left[\begin{array}{ccc}1 & 0 & 0 \\ 0 & 1 & 0 \\ \mathrm{a} & \mathrm{b} & -1\end{array}\right]$ and I is the unit matrix of order 3, then $\mathrm{A}^2+2 \mathrm{A}^4+4 \mathrm{A}^6$ is equal to

(A) $7 \mathrm{A}^8$

(B) $7 \mathrm{A}^7$

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(C) $8\mathrm{I}$

(D) 6

Q.5

If $\mathrm{A}=[a_{ij}]_{n\times n}$ be a diagonal matrix with diagonal element all different and $\mathrm{B}=[b_{ij}]_{n\times n}$ be some another matrix. Let $\mathrm{AB}=[cij]_{n\times n}$ then c_{ij} is equal to

(A) $a_{jj}b_{ij}$

(B) $a_{ii}b_{ij}$

(C) $a_{ij}b_{ij}$

(D) $a_{ij}b_{ji}$

Q.6

If $\mathrm{A}=\begin{bmatrix}1 & 0\\ -1 & 7\end{bmatrix}$ and $\mathrm{I}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$, then the value of k so that $\mathrm{A}^2=8\ \mathrm{A}+\mathrm{kI}$ is

(A) $\mathrm{k}=7$

(B) $\mathrm{k}=-7$

(C) $\mathrm{k}=0$

(D) None of these

Q.7

If $A=\begin{bmatrix}a & b\\ b & a\end{bmatrix}$ and $A^2=\begin{bmatrix}\alpha & \beta\\ \beta & \alpha\end{bmatrix}$, then

(A) $\alpha=2ab,\beta=a^2+b^2$

(B) $\alpha=a^2+b^2,\beta=ab$

(C) $\alpha=a^2+b^2,\beta=2ab$

(D) $\alpha=a^2+b^2,\beta=a^2-b^2$

Q.8

If $\mathrm{AB}=\mathrm{O}$, then for the matrices
$A=\begin{bmatrix}\cos^2\theta & \cos\theta\sin\theta\\ \cos\theta\sin\theta & \sin^2\theta\end{bmatrix}$ and
$\mathrm{B}=\begin{bmatrix}\cos^2\phi & \cos\phi\sin\phi\\ \cos\phi\sin\phi & \sin^2\phi\end{bmatrix},\theta-\phi\text{is}$

(A) an odd number of $\frac{\pi}{2}$

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(B) an odd multiple of π

(C) an even multiple of $\frac{\pi}{2}$

(D) 0

Q.9

The number of all possible matrices of order 3×3 with each entry 0 or 1 is

(A) 18

(B) 512

(C) 81

(D) None of these

Q.10

If number of elements is 20 then how many different types of matrices can be formed if number of rows is always even?

(A) 3

(B) 4

(C) 5

(D) 6

Q.11

Let $A=\left[\begin{array}{ll}1 & 2 \\ 3 & 4\end{array}\right]$ and $B=\left[\begin{array}{ll}a & 0 \\ 0 & b\end{array}\right]$ where a, b are natural numbers, then which one of the following is correct?

(A) There exist more than one but finite number of B 's such that $AB = BA$

(B) There exists exactly one B such that $AB = BA$

(C) There exist infinitely many B 's such that $AB = BA$

(D) There cannot exist any B such that $AB = BA$

Q.12

If B, C are square matrices of order n and if $A = B + C, \mathrm{BC} = \mathrm{CB}, \mathrm{C}^2 = 0$, then for any positive integer N, $\mathrm{A}^{\mathrm{N}+1} = \mathrm{B}^{\mathrm{K}}[\mathrm{B} + (\mathrm{N}+1)\mathrm{C}]$, then K/N is

(A) 1

(B) $1/2$

(C) 2

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(D) None of these

Q.13

If A and B are square matrices of size $n \times n$ such that $A^2 - B^2 = (A - B)(A + B)$, then which of the following will be always true?

(A) $A = B$

(B) $AB = BA$

(C) either of A or B is a zero matrix

(D) either of A or B is identity matrix

Q.14

If $A = \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$, then $A.\,\text{adj}(A) =$

(A) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

(D) $\begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}$

Q.15

If $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, $\mathrm{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $\mathrm{Q} = \mathrm{PAP}^{\mathrm{T}}$, then P $\left(Q^{2005}\right) P^T$ equal to

(A) $\begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} \sqrt{3}/2 & 2005 \\ 1 & 0 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 2005 \\ \sqrt{3}/2 & 1 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & \sqrt{3}/2 \\ 0 & 2005 \end{bmatrix}$

Answers & Solutions

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Q.1 Answer:

$\theta = 2n\pi + \frac{\pi}{3}, n \in Z$

Solution:

We have, $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

$\therefore A^T = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$

Now, $A^T + A = I_2$ **(given)**

$\Rightarrow \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} + \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 2\cos\theta & 0 \\ 0 & 2\cos\theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\Rightarrow 2\cos\theta = 1 \Rightarrow \cos\theta = \frac{1}{2}$

$\Rightarrow \theta = 2n\pi + \frac{\pi}{3}, n \in Z$

Q.2 Answer:

-1

Solution:

Let $B = I + A + A^2 + \ldots + A^{k-1}$

Now multiply both sides by $(I - A)$**, we get** $B(I - A) = \left(I + A + A^2 + \ldots + A^{k-1}\right)(I - A)$

$= I - A + A - A^2 + A^2 - A^3 + \ldots - A^{k-1} + A^{k-1} - A^k$

$= I - A^k = I$**, since** $A^k = 0 \Rightarrow B = (I - A)^{-1}$

Hence $(I - A)^{-1} = I + A + A^2 + \ldots + A^{k-1}$

Thus $p = -1$

Q.3 Answer:

$\left(A^T\right)^2 = A^T$

Solution:

Let A **and** B **be two matrices such that** $AB = A$ **and** $BA = B$ **Now, consider** $AB = A$ **Take Transpose on both side** $(AB)^T = A^T$

$\Rightarrow A^T = B^T \cdot A^T$ **...(1)**

Now, $BA = B$

Take, Transpose on both side $(BA)^T = B^T$

$\Rightarrow \quad B^T = A^T \cdot B^T \quad \ldots(2)$

Now, from equation (1) and (2). we have $A^T = \left(A^T.B^T\right)A^T$

$A^T = A^T\left(B^T A^T\right)$

$= A^T(AB)^T \left(\because (AB)^T = B^T = B^T A^T\right)$

$= A^T \cdot A^T$

Thus, $A^T = \left(A^T\right)^2$

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Q.4 Answer:

$7\ \mathrm{A}^8$

Solution:

$$\mathrm{A}^2=\begin{bmatrix}1&0&0\\0&1&0\\a&b&-1\end{bmatrix}\begin{bmatrix}1&0&0\\0&1&0\\a&b&-1\end{bmatrix}=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$$

$$\mathrm{A}^2=\mathrm{A}^4=\mathrm{A}^6=\mathrm{I}_3\Rightarrow \mathrm{A}^2+2\ \mathrm{A}^4+4\ \mathrm{A}^6$$

$$=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}+\begin{bmatrix}2&0&0\\0&2&0\\0&0&2\end{bmatrix}+\begin{bmatrix}4&0&0\\0&4&0\\0&0&4\end{bmatrix}$$

$$=\begin{bmatrix}7&0&0\\0&7&0\\0&0&7\end{bmatrix}=7\mathrm{I}_3=7\ \mathrm{A}^8$$

Q.5 Answer:

$a_{ii}b_{ij}$

Solution:

$c_{ij}=\sum_{k=1}^{n}a_{ik}b_{kj}$ **(In general) and in a diagonal matrix non-diagonal elements are zero ie,** $a_{ij}=\begin{cases}0, & \text{if } i\neq j\\ a_{ii}, & \text{if } i=j\end{cases}$

So, $c_{ij}=a_{ii}b_{ij}$

Q.6 Answer:

$\mathrm{k}=-7$

Solution:

We have, $\mathrm{A}^2=\begin{bmatrix}1&0\\-1&7\end{bmatrix}\begin{bmatrix}1&0\\-1&7\end{bmatrix}=\begin{bmatrix}1&0\\-8&49\end{bmatrix}$

and $8\ \mathrm{A}+\mathrm{kI}=8\begin{bmatrix}1&0\\-1&7\end{bmatrix}+\mathrm{k}\begin{bmatrix}1&0\\0&1\end{bmatrix}$

$$=\begin{bmatrix}8&0\\-8&56\end{bmatrix}+\begin{bmatrix}\mathrm{k}&0\\0&\mathrm{k}\end{bmatrix}=\begin{bmatrix}8+\mathrm{k}&0\\-8&56+\mathrm{k}\end{bmatrix}$$

Thus, $\mathrm{A}^2=8\ \mathrm{A}+\mathrm{kI}\Rightarrow\begin{bmatrix}1&0\\-8&49\end{bmatrix}=\begin{bmatrix}8+\mathrm{k}&0\\-8&56+\mathrm{k}\end{bmatrix}$

$\Rightarrow 1=8+\mathrm{k}$ **and** $56+\mathrm{k}=49\quad\Rightarrow\mathrm{k}=-7$

Q.7 Answer:

$\alpha=a^2+b^2,\beta=2ab$

Solution:

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$$A^2 = \begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix} = \begin{bmatrix} a & b \\ b & a \end{bmatrix} \begin{bmatrix} a & b \\ b & a \end{bmatrix}$$
$$= \begin{bmatrix} a^2+b^2 & 2ab \\ 2ab & a^2+b^2 \end{bmatrix}; \alpha = a^2+b^2; \beta = 2ab$$

Q.8 Answer:

an odd number of $\frac{\pi}{2}$

Solution:

We have, $AB = \begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix} \begin{bmatrix} \cos^2\phi & \cos\phi\sin\phi \\ \cos\phi\sin\phi & \sin^2\phi \end{bmatrix}$

$$= \begin{bmatrix} \cos^2\theta\cos^2\phi + \cos\theta\cos\phi\sin\theta\sin\phi & \cos^2\theta\cos\phi\sin\phi + \cos\theta\sin\theta\sin^2\phi \\ \cos\theta\sin\theta\cos^2\phi + \sin^2\theta\cos\phi\sin\phi & \cos\theta\cos\phi\sin\theta\sin\phi + \sin^2\theta\sin^2\phi \end{bmatrix}$$
$$= \cos(\theta-\phi) \begin{bmatrix} \cos\theta\cos\phi & \cos\theta\sin\phi \\ \sin\theta\cos\phi & \sin\theta\sin\phi \end{bmatrix}$$

Since $\mathrm{AB} = 0, \therefore \cos(\theta-\phi) = 0$

$\therefore \theta - \phi$ **is an odd multiple of** $\frac{\pi}{2}$

Q.9 Answer:

512

Solution:

There are in total 9 entries and each entry can be selected in exactly 2 ways. Hence, the total number of all possible matrices of the given type is 2^9**.**

Q.10 Answer:

4

Solution:

$$\because 20 = 1\times 20 = 2\times 10 = 4\times 5 = 5\times 4$$
$$= 10\times 2 = 20\times 1$$

Out of this number of rows is even so possible cases are $2\times 10, 4\times 5, 10\times 2$ **and** 20×1**. So number of matrices are** 4**.**

Q.11 Answer:

There exist infinitely many B 's such that $AB = BA$

Solution:

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} = \begin{bmatrix} a & 2b \\ 3a & 4b \end{bmatrix}$$

and $BA = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} a & 2a \\ 3b & 4b \end{bmatrix}$

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If $AB = BA \Rightarrow \begin{bmatrix} a & 2b \\ 3a & 4b \end{bmatrix} = \begin{bmatrix} a & 2a \\ 3b & 4b \end{bmatrix} \Rightarrow a = b$

From the above it is clear that there exist infinitely many B's such that $\mathrm{AB} = \mathrm{BA}$.

Q.12 Answer:

1

Solution:

We have, $\mathrm{BC} = \mathrm{CB}$, **and** $\mathrm{A}^{\mathrm{N}+1} = (\mathrm{B}+\mathrm{C})^{\mathrm{N}+1}$

$= {}^{\mathrm{N}+1}\mathrm{C}_0\ \mathrm{B}^{\mathrm{N}+1} + {}^{\mathrm{N}+1}\mathrm{C}_1\ \mathrm{B}^{\mathrm{N}}\mathrm{C} + {}^{\mathrm{N}+1}\mathrm{C}_2\ \mathrm{B}^{\mathrm{N}-1}\mathrm{C}^2 +$

$\ldots + {}^{\mathrm{N}+1}\mathrm{C}_{\mathrm{r}}\mathrm{B}^{\mathrm{N}+1-\mathrm{r}}\mathrm{C}_{\mathrm{r}} + \ldots$

But given that $\mathrm{C}^2 = 0 \Rightarrow \mathrm{C}^3 = \mathrm{C}^4 = \ldots. = \mathrm{C}^{\mathrm{r}} = 0$

Hence, $\mathrm{A}^{\mathrm{N}+1} = {}^{\mathrm{N}+1}\mathrm{C}_0\mathrm{B}^{\mathrm{N}+1} + {}^{\mathrm{N}+1}\mathrm{C}_1\ \mathrm{B}^{\mathrm{N}}\mathrm{C}$

$= \mathrm{B}^{\mathrm{N}+1} + (\mathrm{N}+1)\mathrm{B}^{\mathrm{N}}\mathrm{C} = \mathrm{B}^{\mathrm{N}}[\mathrm{B} + (\mathrm{N}+1)\mathrm{C}]$

Thus $\mathrm{K} = \mathrm{N}$

Q.13 Answer:

$AB = BA$

Solution:

$A^2 - B^2 = (A-B)(A+B)$

$A^2 - B^2 = A^2 + AB - BA - B^2 \Rightarrow AB = BA$

Q.14 Answer:

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Solution:

A. $(\mathrm{adj}(A)) = |A|I$ **Here** $|A| = \cos^2 x + \sin^2 x = 1$. **Hence,** $A \cdot (\mathrm{adj}(A)) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Q.15 Answer:

$\begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$

Solution:

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Given $Q = PAP^T$

$$\Rightarrow P^TQ = AP^T, \left(\because PP^T = I\right)$$

$$\Rightarrow P^TQ^{2005}P = AP^TQ^{2004}P = AP^TQ^{2003}PA$$

$$\left(\because Q = PAP^T \Rightarrow QP = PA\right)$$

$$= AP^TQ^{2002}PA^2 = AP^TPA^{2004}$$

$$= AIA^{2004} = A^{2005} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^{2005} = \begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$$

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